

HIGH GUARD TO T4 CONVERSION

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Here are my rules for the conversion of *High Guard* USPs to the new Universal Ship Profiles in *Marc Miller's Traveller* (T4). There is also some discussion of the armor conversion rules, with alternative systems proposed.

The advantages to using *High Guard* (and *Book 2*) to design starships are: 1) it's easier than any of the methods currently in print; 2) you can actually easily design ships that are compatible in price and capability with the ships listed in the T4 rulebook.

I have been lobbying for a *High Guard*-like ship design system for nearly a year now. If you ever plan to resurrect *High Guard* (or anything like it), Marc, feel free to use this system or any system based on or inspired by it. It's all yours.

TONS

The T4 tonnage is the same as the *High Guard* tonnage. Take the Tonnage USP value directly from the *High Guard* USP.

HULL TYPE

For the hull type, use the following table to go from the *High Guard* configuration factor to the T4 USP:

HG Configuration	T4 USP
1	Needle/AF
2	Wedge/AF
3	Cylinder/SL
4	Box/SL
5	Sphere/SL
6	Dome/SL
7	Close Structure/No SL
8-9	Planetoid/No SL

VOLUME

Volume is equal to tonnage times 14. $V = 14 \times T$.

COST IN MCr

Same for T4 as in *High Guard*.

CREW

Same for T4 as in *High Guard*.

PASSENGERS HIGH/MEDIUM

Same for T4 as in *High Guard*.

PASSENGERS LOW

Same for T4 as in *High Guard*.

CARGO

Should be listed in tons; same for T4 as in *High Guard*. If you list it in cubic meters instead, then multiply the *High Guard* USP for cargo by 14 to get the T4 USP.

CONTROLS

For starships, Mil Std/Bridge. For small craft, Mil Std or Mil Std/Bridge if you include a small craft bridge. If the *High Guard* computer USP is A, B, C, D, E, F, G, H or J, then add "/fib" to the T4 USP.

TECH LEVEL

Same for T4 as in *High Guard*.

SIZE RATING

Use the table below to convert from the *High Guard* size factor to the T4 USP.

HG Size	T4 Size
0	7
1-9	8
A-J	9
K-Q	10
Q-X	11
Y	12

JUMP RATING

Same for T4 as in *High Guard*.

G RATING THRUSTER PLATES

Same for T4 as in *High Guard*.

POWER PLANT RATING

Same for T4 as in *High Guard*. To get the power plant rating in MW, multiply the power-plant USP by the ship's tonnage, and multiply that by 2.5. Power in MW = $2.5 \times \text{Tonnage} \times \text{PP}$.

FIRE CONTROL RATING and SENSOR RATINGS

Use the following table to convert *High Guard* Computer factors to T4.

HG Computer Factor	T4 Fire Control USP	T4 Sensor USP		
		A	P	J
0	0	0	0	0
1	1	2	2	2
2	2	4	4	4
3	3	6	6	6
4	4	8	8	8
5	5	10	10	10
6	6	12	12	12
7	7	14	14	14
8	8	16	16	16
9	9	18	18	18
A	1	2	2	4
B	2	4	4	8
C	3	6	6	12
D	4	8	8	16
E	5	10	10	20
F	6	12	12	24
G	7	14	14	28
H	8	16	16	32
J	9	18	18	36

R	1	2	4	2
S	2	4	8	4

STRUCTURE

Get the T4 structure USP from the *High Guard* tonnage using the following table:

HG Tonnage	T4 Structure
0+	0
3+	1
7+	2
13+	3
20+	4
27+	5
33+	6
42+	7
50+	8
67+	9
83+	10
125+	11
167+	12
208+	13
250+	14
292+	15
333+	16
417+	17
500+	18
583+	19
667+	20
750+	21
833+	22
1,000+	23
1,167+	24
1,333+	25
1,500+	26
1,667+	27
1,833+	28
2,167+	29
2,500+	30
2,833+	31
3,167+	32
3,500+	33
3,833+	34
4,500+	35
5,167+	36
5,833+	37
6,500+	38
7,167+	39
7,833+	40
9,167+	41
10,500+	42
11,833+	43
13,167+	44
14,500+	45
15,833+	46
18,500+	47
21,167+	48
23,833+	49
26,500+	50
29,167+	51
31,833+	52
37,167+	53
42,500+	54
47,833+	55

53,167+	56
58,500+	57
63,833+	58
74,500+	59
85,167+	60
95,833+	61
106,500+	62
117,167+	63
127,833+	64
149,167+	65
170,500+	66
191,833+	67
213,167+	68
234,500+	69
255,833+	70
298,500+	71
341,167+	72
383,833+	73
426,500+	74
469,167+	75
511,833+	76
597,167+	77
682,500+	78
767,833+	79
853,167+	80
938,500+	81

LASER BATTERIES

Use the table below to convert from *High Guard* laser-battery factors to T4 laser-battery USPs. Two T4 laser-battery USPs are given, one for beam lasers and another for pulse lasers.

HG Factor	T4 Pulse Lasers	T4 Beam Lasers
1	2-1-0-0	1-1-1-1
2	3-1-0-0	2-2-2-2
3	4-2-1-0	3-3-3-3
4	5-2-1-0	4-4-4-4
5	6-3-1-0	5-5-5-5
6	7-3-1-0	6-6-6-6
7	8-4-2-1	7-7-7-7
8	9-4-2-1	8-8-8-8
9	10-5-2-1	9-9-9-9

PLASMA and FUSION GUN BATTERIES

Use the table below to convert *High Guard* plasma or fusion gun batteries to T4 battery USPs.

HG Factor	T4 USP
1	3-0-0-0
2	4-0-0-0
3	5-0-0-0
4	6-0-0-0
5	7-0-0-0
6	8-0-0-0
7	9-0-0-0
8	10-0-0-0
9	11-0-0-0

MESON GUNS and PARTICLE ACCELERATORS

Use the table below to convert *High Guard* factors to the T4 battery USPs for meson guns and particle accelerators. These conversions were made by using the relationship between the *High Guard* size factors, which are linked to weapon factors, and the T4 structure points.

HG Factor	Meson Gun	Particle Accelerator
1	1-0-0-0	1-1-1-1
2	2-0-0-0	2-2-2-2
3	3-1-0-0	3-3-3-3
4	4-2-0-0	4-4-4-4
5	5-3-1-0	5-5-5-5
6	6-4-2-0	6-6-6-6
7	7-5-3-1	7-7-7-7
8	8-6-4-2	8-8-8-8
9	9-7-5-3	9-9-9-9
A	13-10-7-5	13-13-13-13
B	19-16-13-10	19-19-19-19
C	24-21-18-15	24-24-24-24
D	25-22-19-16	25-25-25-25
E	26-23-20-17	26-26-26-26
F	28-25-22-19	28-28-28-28
G	29-26-23-20	29-29-29-29
H	29-27-24-21	29-29-29-29
J	31-28-25-22	31-31-31-31
K	32-29-26-23	32-32-32-32
L	38-35-32-29	38-38-38-38
M	42-38-34-30	42-42-42-42
N	44-40-36-32	44-44-44-44
P	46-42-38-34	46-46-46-46
Q	50-45-40-35	50-50-50-50
R	52-47-42-37	52-52-52-52
S	58-52-46-40	58-58-58-58
T	62-56-50-44	62-62-62-62

SANDCASTER BATTERIES

Use the table below to convert from the *High Guard* factor to the T4 USP. The T4 USP is given for 3 tech levels.

HG Factor	TL7	TL8	TL10
0	0	0	0
1	1	—	—
2	3	1	—
3	6	3	1
4	8	6	3
5	10	8	6
6	20	10	8
7	30	20	10
8	—	30	20
9	—	—	30

MISSILE BATTERIES

Use the table below to get the number of missiles that can be controlled in flight by a battery at a single time from the *High Guard* factor. The T4 USP is given for two different tech levels.

HG Factor	TL7	TL13
0	0	0
1	1	—
2	3	1
3	6	3
4	12	6
5	18	12
6	30	18
7	60	30
8	120	60
9	180	120

FORCE FIELDS

The flicker rate of the black globe is equal to the *High Guard* force-field factor times 10%.

NUCLEAR DAMPERS

The nuclear damping screens reduce each nuclear missile salvo by the following amount:

HG Factor	T4 USP
0	0
1	1
2	3
3	6
4	12
5	18
6	30
7	60
8	120
9	180

REPULSOR BATTERIES

Each repulsor battery reduces one missile salvo by the following amount:

HG Factor	T4 USP
1	30
2	60
3	120
4	180
5	300
6	600
7	1200
8	1800
9	3000

MESON SCREENS

HG Factor	T4 USP
1	9
2	13
3	19
4	24
5	25
6	26
7	28
8	29
9	30

ARMOR

I present below three schemes for converting *High Guard* armor values to those of T4. Scheme 'A' is my recommended, simple, arbitrary scheme. The biggest weakness is that it can lead to abusively powerful small craft designs. It works well for small ships with small armor values, big ships with high armor values, and medium sized ships with medium-sized armor values.

Scheme 'B' is my interpolation of an "official" (but nonsensical) conversion detailed in *Striker*, which relates *High Guard* armor values to a single hull thickness. It works well for ships in the range of 500-2000 tons.

Scheme 'C' is based on Guy Garnett's take on the official conversion, which he explained on the Traveller Mailing

List in 1997; here I did not interpolate between the points. It works well for ships around 100,000 displacement tons.

HG Factor	A	B	C
0	0	17	10
1	10	23	70
2	20	26	80
3	30	30	90
4	40	34	100
5	50	38	100
6	60	44	110
7	70	47	110
8	80	50	110
9	90	54	110
10	100	58	120
11	110	61	120
12	120	65	130
13	130	69	130
14	140	74	140
15	150	79	140
16	160	82	150
17	170	85	150
18	180	88	160
19	190	92	160
20	200	95	170
21	210	99	170

Of all these *High Guard* to T4 armor conversions, I recommend scheme 'A', above. It is the simplest, and it works fairly well for most ships; it's good enough for government work, and better than good enough for games.

However, below is a scheme based on equalizing the displacement of the armor in both *High Guard* and T4. It is the most "realistic" scheme (as if "realistic" meant anything in this context) since it is based on physical and geometrical principles. Unfortunately, the equation giving the relation between the *High Guard* and T4 USPs is a real monster. Here it is for *High Guard* armor factors equal to 15 or less:

$$T4 = 4.7 \times (M \times 14 \times 1000000 \div 3.14 \times 3 \div 4)^{(1 \div 3)} \div 3 \times (HG + 1) \div 100$$

M = ship's displacement in tons.

HG = *High Guard* armor USP rating.

T4 = armor rated as centimeter's thickness of hard steel.

After using this equation, take 'T4' and use that as the Actual Value on the USP Conversion Chart on page 107 of *Starships*, look up the USP Rating, and then multiply that value by 10. (And people wonder why I find the new ship-design rules so complicated!)

Obviously, the above formulation is not amenable to a simple look-up table, but I did my best. To use the lookup table below, find the column for the hull tonnage closest to your ship's tonnage. Go down the column till you match your ship's *High Guard* armor rating (in the leftmost column). The number that you are referencing is the T4 armor value.

HG Factor	Hull Tonnages					
	10	100	1,000	10,000	100,000	million
0	0	0	0	0	0	0
1	0	10	20	30	60	90
2	0	10	20	40	80	100
3	10	20	30	60	90	110
4	10	20	30	70	100	120
5	10	20	40	80	100	130
6	10	20	50	80	110	140
7	20	30	50	90	110	150
8	20	30	60	90	110	160
9	20	30	60	100	120	160
10	20	30	70	100	120	170
11	20	40	70	100	130	170
12	20	40	80	100	130	180
13	20	40	80	100	140	180
14	20	50	80	110	140	190
15	30	50	80	110	140	190
16	40	80	100	140	180	240
17	50	80	100	140	180	240
18	50	80	110	140	190	240
19	50	80	110	140	190	240
20	50	80	110	150	190	250
21	50	80	110	150	190	250